

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly

Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

ANSWERS

Question 1: Smart Campus Subnet Allocation (192.168.100.0/24)

- Step 1: Calculate the subnets
- School of Computing → /25
 - School of Electronics → /26
 - School of Mechanical Engineering → /27
- Step 2: Determine network and broadcast addresses

Department	Network Address	Broadcast Address
Computing	192.168.100.0	192.168.100.127
Electronics	192.168.100.128	192.168.100.191
Mechanical	192.168.100.192	192.168.100.223

- Step 3: Find the valid host ranges
- Computing: 192.168.100.1 – 192.168.100.126
 - Electronics: 192.168.100.129 – 192.168.100.190
 - Mechanical: 192.168.100.193 – 192.168.100.222
- Step 4: Calculate usable hosts
- Formula: 2^(32–Prefix) – 2
- /25 → 126 hosts
 - /26 → 62 hosts
 - /27 → 30 hosts
- Step 5: Identify the unused IP range
- Unused range: 192.168.100.224 – 192.168.100.255
- Final Answer: The network is divided into three subnets with 126, 62, and 30 usable hosts, and the remaining addresses are 192.168.100.224–255.

Question 2: Company Network Subnetting (10.10.0.0/16)

- Step 1: Allocate subnets
- Software Development → /24
 - Quality Assurance → /25
 - Human Resources → /26

- Executive Management → /27

Step 2: Determine network and broadcast addresses

	Department	Network	Broadcast
	Software	10.10.0.0	10.10.0.255
	QA	10.10.1.0	10.10.1.127
	HR	10.10.1.128	10.10.1.191
	Executive	10.10.1.192	10.10.1.223

Step 3: Identify host ranges

- Software: 10.10.0.1 – 10.10.0.254
- QA: 10.10.1.1 – 10.10.1.126
- HR: 10.10.1.129 – 10.10.1.190
- Executive: 10.10.1.193 – 10.10.1.222

Step 4: Calculate usable hosts

- /24 → 254
- /25 → 126
- /26 → 62
- /27 → 30

Step 5: Find the unused IP range

Unused range: 10.10.1.224 – 10.10.255.255

Final Answer: The company network supports 254, 126, 62, and 30 hosts, and the remaining address space starts from 10.10.1.224.

Question 3: IPv6 Addressing

Given: 2001:0db8:abcd:0000:0000:0000:0000/64

Step 1: Compress the IPv6 address

Compressed form:

2001:db8:abcd::/64

Step 2: Expand the compressed address

Expanded form:

2001:0db8:abcd:0000:0000:0000:0000/64

Step 3: Identify the network prefix

With /64, the network prefix is:

2001:db8:abcd::/64

Step 4: Calculate host addresses

Formula:

$$2^{(128-64)} = 2^{64}$$

Step 5: Final host capacity

total host addresses:

18,446,744,073,709,551,616

Final Answer: The compressed address is 2001:db8:abcd::/64, and the subnet supports 2^{64} host addresses.

Question 4: Routing Decision

Routing Table:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP: 192.168.2.45

Step 1: Determine the destination subnet

IP 192.168.2.45 belongs to:

192.168.2.0/24

Step 2: Find the matching route

Matching entry:

192.168.2.0/24

Step 3: Determine the next hop

The packet is forwarded through the interface connected to the 192.168.2.0/24 network.

Step 4: Check for default routing

If no route matches, the router uses the default route.

Step 5: Final routing result

- Destination subnet: 192.168.2.0/24
- Matching route: 192.168.2.0/24
- Next hop: Interface to 192.168.2.0/24
- Default route: 0.0.0.0/0

Final Answer: The packet is forwarded to the 192.168.2.0/24 network, and unmatched packets use the 0.0.0.0/0 default route.

Question 5: Two Office LANs

Administration LAN: 192.168.10.0/26

Finance LAN: 192.168.10.64/26

Step 1: Determine network and broadcast addresses

LAN	Network	Broadcast
Administration	192.168.10.0	192.168.10.63
Finance	192.168.10.64	192.168.10.127

Step 2: Identify host ranges

- Administration: 192.168.10.1 – 192.168.10.62
- Finance: 192.168.10.65 – 192.168.10.126

Step 3: Assign IPs to three computers

Administration:

- PC1 → 192.168.10.2
- PC2 → 192.168.10.3
- PC3 → 192.168.10.4

Finance:

- PC1 → 192.168.10.66
- PC2 → 192.168.10.67
- PC3 → 192.168.10.68

Step 4: Identify default gateways

- Administration Gateway: 192.168.10.1
- Finance Gateway: 192.168.10.65

Step 5: Final network configuration

All Administration hosts use 192.168.10.1 as the gateway, and all Finance hosts use 192.168.10.65.

Final Answer: The two /26 LANs have separate network and broadcast addresses, valid host ranges, assigned computer IPs, and default gateways 192.168.10.1 and 192.168.10.65.